Your text here 1

8. NMAP to Discover Live host using nmap Scans (ARP, ICMP, TCP/UDP) on Try hack platforms 8

9. Implementation of Subnetting, CISCO Packet Tracer Simulator. 8

10 Interworking with router in CISCO Packet tracer simulator 8

11 Routing at Network layer 9

12 End to End Communication at transport Layer 9

13 Implement your own ping program 9

14 Write a code using RAWSockets to implement packet Sniffing. 10

15. Analyze various types of Servers using tool. 7

Answered
[2-Apr-26]

Exp 01

Study of various Network Commands Used in Linux and windows

Aim:- To study of various network commands used in linux and windows

windows Commands :-

* arp -a
  - ARP is short form of address resolution protocol it will show ip address of your computer along with ip address and MAC address of your router

* hostname
  This is the simplest of all TCP/IP Commands. It displays name of your computer

* ip config/all:
  This command display detailed Config information about your TCP/IP connection.

* nbtstation :-
  The Command helps solve problems with netbios name resolution.

Linux Commands

* ip

The ip command an show address information manipulate routing plus display network various devices interfaces and tunnels.

* [root@ Search] # ip address show to show the ip address assigned to an interface

* [root@ Server] # ip address add 192.168.1254 dev enp502 to assign a ip to an interface

* config

The of config command is a staple in many Sysadmin tool belt for configuring and troubleshooting networks

* [root@server] # mtr google.com.

the mtr command show you the statistics

* [root@serverN] # mtr -y google.com show numeric ip address

* [root@serverN] # mtr -b google.com show numeric ip address and host names.

* tcp dump It is designed for capturing and displaying pachets

* [root@serve] # dnf install -y tcp dump

* [root@serverN] # tcpdump -i

* For traffic comming from 8.8.8.8 use

[root@serverN] # tcpdump -i eth0 -c10 host 8.8.8.8

* For Specific host -

[root@Server ~] # tcpdump - path host 8.8.8.8 and port 53

* To Capture only HTTPS traffic

[root@Server ~] # tcp dump - p eth0 - c 10 host www.google.com and port 443

* ping is a tools that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Messaging protocol (ICMP).

* To ping a host ten times use the following command

Result :-

The above commands are executed Successfully.

Experiment-02

Studying of different types of network Cables

Aim:-

To study of different types of network cables.

a understand different type of network cable different types of cables of network cables.

* Unshielded twisted pair (UTP) Cable. different types
* Shielded Twisted pair (STP) cable.
* Coaxial cable
* Fibre optic cable.

Cable type	category	maximum data	Advantage
UTP.	category 3	10 bps	Cheaper in cost
	category 5	upto 100 Mbps	Easy to install as they have smaller overall diameter.

Category 5e	1Gbps	More prone to (EMI) electromagnetic interference & noise.
Category 6a	10Gbps	* Shielded * Faster than Vip
Category 7	10Gbps	* Expensive * Greater installation effort
RG-6 RG-59 RG-11	10-100 Mbps	* High bandwidth * Low loss bandwidth Disadvantage * Limited distance * Cost * SPZ is bulky

Single module

Multi module

100 Gbps

Advantage

* High speed
* Heigh bandwidth
* Heigh security
* long distance

Results

The above different types of Cables are explained successfully.

Ex - 03

To Study the Packet tracer tool

Installation and user Interface

Aim:

To Understand environment of Cisco Packet tracer to design simple network

Introduction:

As Simulator as the name suggest Simulates network devices and it environment

→ 1. It allows you to model computer system without the need for dedicated equipment

→ 2. It help you to prott your network configuration.

→ 3. It is available for the Linux and Windows desktop environment

Installing packet tracer

* After downloading the appropriate Version for os, the installion can be Completed by following steps

Exp-04

Setup and Configure a LAN using a Switch and Ethernet Cables in your lab.

Aim:

To Setup and Configure a LAN Using a Switch and ethernet Cables your lab.

LAN - A local area network refers to a network that connect devices within a limited area, Such as an office building, School, or home

Setup LAN

Step1:

↳ Plan and design an appropriate network topology and location

Step2:

↳ You can tables 4 computers Switch and 4 ethernet tables.

Step3:-

Connect your Computer to network Switch via ethernet Cable.

Step 4:-

Assign IP address to your P's.

Step 5:-

Configure a network Switch

1.) Connect your Computer

2.) Log-in to the web interface

3.) Configure basic setting

4.) Assign Ip address

Step 6:-

Check Connectivity between Switch & other machine by using ping Command.

Step 7:-

Select folder -> go to properties Click Sharing tab -> Share in with everyone. One

Step 8:

Try to access the Shared folder from other computers of the network.

LAN Connection

Result:

Thus the Configuration of a LAN using a Switched and Ethernet cables are Successfully performed.

Exp-05

Experiment packet capture tool & Wire Shake

Aim:

To capirments an packet

Capture tool : Wire Shark

* Sniff message being sent/received by your computer

* Store and display the contents of various protocol fields in message

* Passive program

i) Never sends packet itself

ii) No packets add to it.

iii) Receive a copy of all packet (sent / receive)

Packet Sniffer Structure: Diagnostic tools

* Top dump - Eg 3 cp dep one has 10.12.90.42.w.

* wireshake - wireshark -r lsce.s.out

Procedures

1) Select the local Area network in Wire Shark

ii) Go to capture option

iii) Select capture and Stop automatically after two packets

iv) Then click Start capture

v) Save packets

Create a filter to display only tcp/udp packets. Inspects the packets and provide flow graph.

* Go to Search bar at the top of the tool.

* Search Tcp paches in Search bar.

* To see flow graph click Statistics flow graph

* Creating a filter to display only packets and provide the flow graph

* Creating filter to display tcp only packets and inspect the packet.

Ex 8

Result:
Wireshark help to us in analyzing and understand our network traffic & packet flow throw var filters and flow Graph.

EX 8 6

PYTHON IMPLEMENTATION OF HAMMING CODE CONCEPT

Error correction of Data - Link layer

Hamming code is a set of error-correction codes that can be used to detect and correct errors when data transmitted from sender to receiver

Aim

To implement error detection using Hamming code Technique at data

SENDER Programs

Code 8 0

```
def hamming_encode(data):
    encode = ""
for p in range(0, len(data)-4)
    d = data[p:p+4]
    if len(d) 4:
        d = d.ljust(4-'0')
```

```
d1, d2, d3, dy = map(int, o)
P1 = d1 ^ d2 ^ dy.
P2 = d1 ^ d3 ^ dy
P4 = d2 ^ d3 ^ dy
Code = str(p) + str(p2) + str(d1)
    + str(p4) + str(d2) + str(d3) +
    str(dy)
encode t= code
return encode
text = input("Enter text: ")
binary = ' '.join(format(code))
        ('b0') for c in text)
Print (* Binary data" binary)
encode - data = hamming-Coded (binary)
with open("channel.txt", "w") as file
Print ("data saved to channel.txt")
```

RECEVIER PROGRAM

Code 80

```
def hamming_decode(data):
    corrected = ""
    error_found = False

for i in range(0, len(data), 7):
    block = list(map(int, data[i:i+7]))
    if len(block) < 7:
        continue
    p1, p2, d1, p4, d2, d3, d4 = block
    s1 = p1 ^ d1 ^ d2 ^ d4
    s2 = p2 ^ d1 ^ d3 ^ d4
    s4 = p4 ^ d2 ^ d3 ^ d4
    error_position = s1 + (s2*2) + (s4*4)
    if error_position != 0:
        error_found = True
        actual_position = error_position
print("Error detected at bit position", actual_position)
```

Input:

Enter the text : HELLO

Input:

HELLO

output:

Enter the text : Hello.

Error detected at position : 1

Error corrected

Original Message : Hello.

Results

The above hamming code was executed Successfully.

EX-07

Flow control at data Link Layer

Aim :

To write a program to implement flow control at dataLink layer using sliding window.

Create a sender program with flowing features :

1) Input window size from user
2) Input text message from the user.
3) consider character per frame
4) Create a frame with following field
5) Send the frames.

[Print output on screen and save it in a file called sender buffer)

6) wait for the acknowledgment from the receiver

7) Check ack - reader file called receiver buffer

8) Check Ack field the Acknowledgement

9) If the Acknowledgement numbers as expected send new set of frames according.

Create a receiver with following features as

1) Read a file called sender-Buffer

2) Check frame no

3) if frame no are as expected

write the appropriate Ac, no in Receiver - Buffer file else writ no in receiver - Buffer file.

Sender code as

```
import os
import time

Base-Dir = os.path.dirname(os.path.abs()
```

```
Print ("Sender Buffer", Send.file)
print ("Receiver Buffer", receiver.file)
window_size = int (input ("Enter window
                          Size:"))

message = input ("Enter text message:")
if window size <= 0:
    print ("window Size must be
           Grater than zero")

    False Systemed;

Frames[]
    While Start < total_frames
        end = minstart + window_size,
              total frames)
        print ("Sending frames", start,"
                                to", end)
elif are type == "NACK"
    print ("NACK received")
    print ("Retransmiting from
           frames:", Retno)
```

else.

print ("UNknown escape Retrying Current window :")

Print ("Total frames" total_frames)

Receiver code Input :-

Sender.Buffer.txt

0 H

1

2 E

3 L

L

4 O

Receiver code output :-

Received 0 : H

Received 1 : E

Received 2 : L

Received 3 : L

Received 4 : O

8/10 8/10

Results

Therefore the implement flow control of datalink layer using Sending window protocol executed successfully.

Exp 08

NMAP to discover Live Host using N map

Scans CARP, ICMP, TCP/UDP on Try hackme platform

Aim 8

To Discover Live Host using Nmap Scan CARP, ICMP, TCP/UDP on the try hackme platform Room link:

https://tryhackme.com/room/nmap

Subnetworks 8

A network segment is a group of Computer linked through a shared medium Line an ethernet switch on wifi access point In network, a Subnetwork typically Consist is a Logical connection

124 Subnets The feature a Subnet key to run Compiler i> From Computer 1

ii> TO Computer 1

iii> pack type = "ARP Request"

iv> Data Computer

Discovery Live host

TCP IP Layer

i> ARP from Link Layer

ii> ICMP from Network Layer

iii> Tcp from Transport Layer

iv> UDP from Transport Layer

and a parent with following

i> From Computer 1

ii> to computer 2

NMAP HOST Discovery using ARP.

How Case can determine which are operational?

post Scanning on host that are offline not in use. This allows you to discover live system On Same subnet as you target machine consider any part sans

Result

Therefore the Discover Live Host using Nmap Scan platform Link was executed Successfully

2/15 x 8/60

1X-7

Aim:

Implementation of Subnetting in cisco packet tracer Simulator

Procedure:

i> classical IP Subnetting is a technique that allows for more efficient use of ip address by allowing for subnet masks networks.

ii> Creating a Network topology

The first step in implementing classes IP Subnetting is to Create a network topology in packet track

To create a network technique to add devices

iii> Adding the previous

Once we have created our networking topology, we can add devices in Pt

iv> Subnetting

To Subnet the network address of 192.168.1.0/24 to provide energy you for atleast 5

The Ip addressing for the network show topology

i) Couter R1
ii) Gigabit Element 0/0 : 192: 168 1.1
iii) gigabit Ethernet 0/2 · 192 · 168 · 2-1
iv) Switch S1:
v) Fast Ethernet 0/1: 192: 168: 1.0/27
vi) PC1 : 192.1681: 11
vii) PC2 192: 168·1:12
viii) PC3 192: 168: 1·13
ix) PC4 : 192: 168: 1: 14
x) PC5 : 192: 168: 1·15
xi) Fast Ethernet 0/2: 192·168·2·0/27
xii) FC5 : 192 168 4: 15

Configure the device

enable
Configure terminal
Interface Fast Ethernet 0/0
Ip address {ipaddress} {subnet mask}

no Shutdown
exit
Interface Fast Ethernet 0/1
IP address {IP address} {Subnet mask}
no Shut down
exit

Results:

Therefore the Subnetting in Cisco Packet Tracer Simulator was executed Successfully

10.
a) Internet working routes in cisco packet tracer

Aim
to design and configure a simple internetwork using a router in cisco packet tracer and fast connectivity two pcs

Procedure:

Step 1: Configure router:
* Select the routes and open cli
* press enter to start configure router
* type enable to activate the privilages model.

Step 2:
* Assign ip address to every pc in the network.
* Select the P1, go to desktop and select ip configure and assign ip address default gateway
* Assign's the default gate way of P'cs 192.16.8.10.1

Step 3

* Connect fast ethernet a part of plc with fast ethernet o/o part of router using "cop post Straight through Cable

* Connect fast ethernet a power with past ethernet all port of row as using a copper Straight through Cable

Result:

Therefore, the Internet working router in cisco packet tracer

10>
b> Design and Configure an Internetworking using Wireless router, DHCP server and Internet closed.

Aim:

to design and configure an internetworking using a wireless router DHCP server and Internet cloud in CISCO packet tracer.

procedure

* open cisco packet tracer and Create required network topology
* Connect pc : laptop wireless router, cable, modem, Internet cloud and server
* Configure laptop to the wireless router network and pc to DHCP
* Configure the servers with IP 208.87.220.25 and enable DHCP and Dive
* verify Connecting using IP Configure and ping Cisco.com

11>

a> Simulate static routing configuration using cisco packet tracer

AIM:

To simulate a static routing configure using cisco packet tracer.

Procedure

* open cisco packet tracer and Create the given network topology
* connect 3 routers and PCs using suitable cable
* Connect IP address for router and PCs
* Enable and router interface using no shutdown
* Configure static routers on router gates router 2 using IP mobile
* Set main and backup routes using difference administrative distance values
* Test Connectivity
* User tracer verify the routing Path

* Disconnect the main link and verify that the backup router work

Result:

Therefore, the Simulate State, routing Configuration using CISCO packet tracer was executed successfully

11>

b> Simulate RIP using Cisco packet tracer

Aim:

To Simulate RIP using 3 routers and 2 pc's cisco packet tracer.

Procedure:

* Create the topology with 3 routers and 2 pc's
* Connect the device using Suitable Cables.
* Assign the given IP address to both pc's
* Configure fast ethernet and serial interface for all routers
* Enable interface using no shutdown
* Set clock execute of 11000 and bandwidth in serial interface

* Enable RIP on all router using the CISCO
* set/ clock / execute of 4010 and bandwidth
* Add directory connected network using network Command.
* Disconnected the direct creates track 2 Link and router again to verify RIP select the alternate route

Result:

Thus the Simulation of RIP using router 2pc's CISCO packet trailer has be Successfully done.

12. End - End Communicated at Transport Layer

a) Implement echo client server using TCP / UDP Socket

Implement a chat program using Socket program

Tcp:

```
import socket
import threading
def start-server()
  Server = Socket.Socket(Socket.AF
- INET. Socket. SOCK - Stream)
  Server bind (("127.0.0.1", 5000))
    Server. listen (1)
print ("[SERVER] Listening on 127.0.0.1
:5000")
conn.add.server.accept()
  Printf ("f "[SERVER] Connected by
          folder}")
  while true:
        data = conn.recv(1024).decode
  if not data or data.lower() == 'quit':
    Print (SERVER] Connection Closed
                  by Circuit)
      break
    print (f" (SERVER) Connection
Con. close ()  closed by Circuit)
Server. close()
```

```
# -client - -
    def start_client()
  client = socket.socket (socket
   AF - INET, socket SOCK-Stre
                          am)
  client.connect (('127.0.0.1', 5000)
```

Output:

```
{ server listening on 127.0.0.1 ; 5000 [client]
connected to server
Hello
{ server } received Hello
[server Response] Server received Hello Enter
 usuage (or 'quit' or exit)
```

UD

```
  input socket
  import Threading
  server - -
def udp_server();
    s = socket.socket (socket
      socket Digram)
  s.find ('127.0.0.1', 12000)
  Print ( Server) Running on '127.0.0.1', 12000
```

Result: Thus static counting configuration & RIP is simulated using Cisco Packet Tracer.

Implement own ING Program

13) Aim :

To implement my own Ping Programm

ode :

```
Input as
Inpert platform
Inport Sub proces
Import subprocess
def ping (host "google.com)
    Param = "-n" if platform.system().lower()
            == "windows"
    Command = ["ping", Param. st(count) host]
    output= Subprocess Check - output (comma
                                       nd
            Universal- new line = True)
    Print (output)
except exception-ase :
        print (f" Ping faild= {e}")
if -- name -- == '-- main--"
    Ping ("8.8.8.8")
```

output :

Pinging 8.8.8.8. with 32 bytes of data

Replying from 8.8.8.8: bytes -32 time
= 9ms TTC =

Reply from 8.8.8.8 : bytes=32 time=
Reply from 8.8.8.8 byt 8 ms TTC=(

Ping Statistic for 8.8.8.8

Packets. sent-4 Received-4

Approximate round trip time in milli-Second

minimum = 5ms, Maximum = 9ms, Average = 7ms

Result:

Thus the ping program is written and executed Sucessfully 9/10

Ex-14. Sockets to implement packet Sniffing

Aim:
To write all import ip, Tcp, Udp, Icmp.

```
if ip in packet
    ip_layer = packet[IP]
    Protocol = ip_layer.proto
    Src_ip = ip_layer.src
    dst_ip = ip_layer.dst
    protocol_name = ''
    if protocol==1:
        protocol_name="ICMP"
    elif protocol==6:
        protocol_name="Tcp"
    elif protocol==17:
        protocol_name= Udp
    else:
        protocol_name = "UNKNOWN Protocol"
print(f"protocol: {protocol_name}")
print(f"Source IP: {src_ip}")
print(f"Destination IP: {dst_ip}")
```

Sniff (iface = "wi" pm-packet-callback, filters = "ip"
store = 0)

output :-

C/User/Vishnu/Downloads: python packet.py

protocol: TCP
Source ip : 57.15.141.117
Destination ip : 192.168.16

protocol : TCP
Source ip : 192.168.1.6
Destination ip : 57.153.141.117

protocol : UDP
Source ip : 192.168.1.6
Destination ip : 192.168.1.1

protocol : UDP
Source ip : 192.168.1.1
Destination ip : 192.168.1.6

Result:

This the raw Sockets is used to implement packet Sniffing Successfully.

2/15 x 15/15